

Rahul Patnala

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Professional Summary

AI/ML Enthusiast specializing in **Generative AI, Agentic AI, and LLM orchestration**. Experienced in designing multi-agent LangGraph workflows, Retrieval-Augmented Generation (RAG) pipelines, hybrid retrieval systems, LLM routing, guardrails, and tool-enabled AI applications.

Education

B.Tech in Computer Science and Engineering (AI & ML) , GMR Institute of Technology CGPA: 9.04	2022 – 2026
Intermediate , Sri Viswa Junior College Percentage: 96.8%	2020 – 2022
SSC , Srujana Vidyalaya CGPA: 9.82	2019 – 2020

Technical Skills

Programming: Python, Java, SQL, JavaScript, C, C++
Generative AI: Generative AI, Agentic AI, LLMs, LangGraph, LangChain, RAG, Prompt Engineering, Structured Outputs, LoRA Fine-Tuning, Model Context Protocol (MCP)
Machine Learning: Machine Learning, Deep Learning, NLP, Scikit-Learn, TensorFlow, NumPy, Pandas
Backend: Spring Boot, FastAPI, Flask, REST APIs, Microservices, React, Streamlit
AI Infrastructure: Qdrant, ChromaDB, PostgreSQL, Supabase, Hugging Face, BGE Embeddings, FlashRank
Evaluation & Tools: RAGAS, LangSmith, NVIDIA NeMo Guardrails, Portkey, Git, GitHub

Experience

GenC Intern – Cognizant	Feb 2026 – Jun 2026
ComNet360 Platform: Developing an enterprise communications system using a Spring Boot microservices architecture, featuring an API Gateway, JWT-based RBAC, and automated SLA breach detection. - Building a React-based frontend with role-specific dashboards and integrating multiple backend services through a centralized gateway for real-time incident analytics.	

Projects

TourAlly – Multi-Agent AI Travel Planner	GitHub
- Built a multi-agent LangGraph workflow using a supervisor-specialist architecture to dynamically route requests across Flights, Hotels, Weather, and Budget agents powered by Llama models through Groq. - Integrated real-time information using Model Context Protocol (MCP), AviationStack, Tavily, and OpenWeather APIs with dynamic currency conversion for itinerary cost normalization. - Implemented input-validation guardrails, human-in-the-loop checkpoints, Supabase state persistence, and LangSmith observability for reliable multi-turn agent execution.	
Banking Knowledge Intelligence Platform (BKIP)	GitHub
- Architected a stateful Agentic RAG system using LangGraph and FastAPI for querying banking policies, RBI circulars, and internal guidelines with dynamic intent routing. - Built a hybrid retrieval pipeline using Qdrant, BGE embeddings, and FlashRank ONNX reranking to improve context precision and retrieval quality. - Added NVIDIA NeMo Guardrails, PII protection, LLM fallback routing via Portkey, and RAGAS/LangSmith evaluation for production-oriented reliability and observability.	

Achievements

- 1st Place – DataCraze, Stepcone 2025:** Analyzed a 100K+ Twitter dataset to identify actionable sentiment trends and developed an interactive analytics dashboard. GitHub
- 1st Place – DataViz, Stepcone 2024:** Developed a geographic sales analytics dashboard using feature engineering, custom mapping, and automated business summaries. GitHub

Certifications

- AI & Data Science Program** – DRISHTI CPS, IIT Indore (2025)
- Deep Learning** – NPTEL IIT Ropar (2025), **Top 1% Performer**
- Data Science and Machine Learning Bootcamp** – Udemy (2025)